

Responding to a Zoonotic Disease Outbreak - Epidemiological interpretation and implications for decision-making in One Health

Afternoon hands-on tutorial

2026-07-03

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1 Overview

We will explore surveillance data from Integrated Bite Case Management (IBCM) in MIMAROPA region. IBCM is a One Health approach to rabies surveillance that links post-exposure vaccination of bite victims, investigation of biting animals, community sensitisation and programmatic coordination, to improve prevention, detection and response. IBCM was introduced to Romblon and Oriental Mindoro provinces in 2020, although COVID-19 restrictions prevented full implementation. In 2022 IBCM expanded to Marinduque province and in 2024 to Occidental Mindoro.

Using these data you will examine where outbreaks occurred; how these outbreaks spread; and why. In groups you will look more deeply into one province and consider the implications for control and prevention.

2 Learning objectives

In this session you will:

- 1) gain understanding of how to **critically interpret epidemiological data** considering a **One Health perspective** and the **quality of evidence**.
- 2) In so doing you should: **identify gaps in surveillance and control**; and **make recommendations** for future action.

Code is folded but available via the **Show code** button.

i Files you need (in your workshop folder)

- `dat_all_anon.csv` — human and animal rabies cases for MIMAROPA
- `mdv_summary.csv` — dog vaccination data (timing of vaccination campaigns in the region by province)
- `rolling_vac_cov_yr.csv` — hupreprocessed estimates of vaccination coverage by

municipality

- PHL_municipality.shp (and its .dbf, .shx, .prj files) — municipality polygons for MIMAROPA

i import and process the data for plotting

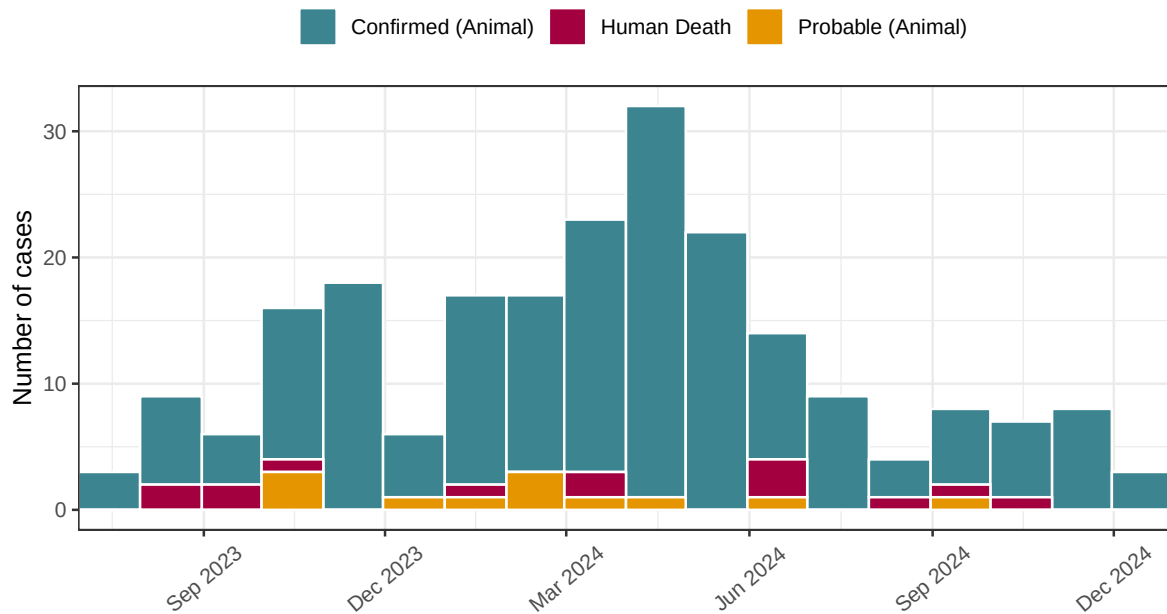
```
# A tibble: 6 x 9
  barangay      municipality province case_status date case_status_fct case_type
  <chr>         <chr>         <chr>    <chr>    <chr> <chr>         <chr>
1 Apitong      Boac           Marindu~ Confirmed 24/0~ Confirmed (Ani~ Animal
2 Balogo       Boac           Marindu~ Confirmed 10/1~ Confirmed (Ani~ Animal
3 Agumaymayan Boac           Marindu~ Confirmed 06/1~ Confirmed (Ani~ Animal
4 Laylay       Boac           Marindu~ Confirmed 23/1~ Confirmed (Ani~ Animal
5 Bangbangalon Boac           Marindu~ Probable 13/1~ Probable (Anim~ Animal
6 Bangbangalon Boac           Marindu~ Probable 16/1~ Probable (Anim~ Animal
# i 2 more variables: year <dbl>, island <chr>

[1] "2019-02-28" "2019-03-31" "2019-04-30" "2019-05-31" "2019-06-30"
[6] "2019-07-31" "2019-08-31" "2019-09-30" "2019-10-31" "2019-11-30"
[11] "2019-12-31" "2020-01-31" "2020-02-29" "2020-03-31" "2020-04-30"
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[21] "2020-10-31" "2020-11-30" "2020-12-31" "2021-01-31" "2021-02-28"
[26] "2021-03-31" "2021-04-30" "2021-05-31" "2021-06-30" "2021-07-31"
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[36] "2022-01-31" "2022-02-28" "2022-03-31" "2022-04-30" "2022-05-31"
[41] "2022-06-30" "2022-07-31" "2022-08-31" "2022-09-30" "2022-10-31"
[46] "2022-11-30" "2022-12-31" "2023-01-31" "2023-02-28" "2023-03-31"
[51] "2023-04-30" "2023-05-31" "2023-06-30" "2023-07-31" "2023-08-31"
[56] "2023-09-30" "2023-10-31" "2023-11-30" "2023-12-31" "2024-01-31"
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[66] "2024-07-31" "2024-08-31" "2024-09-30" "2024-10-31" "2024-11-30"
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[81] "2025-10-31" "2025-11-30" "2025-12-31" "2026-01-31" "2026-02-28"
[86] "2026-03-31" "2026-04-30" "2026-05-31"
```

3 Describing the outbreak:

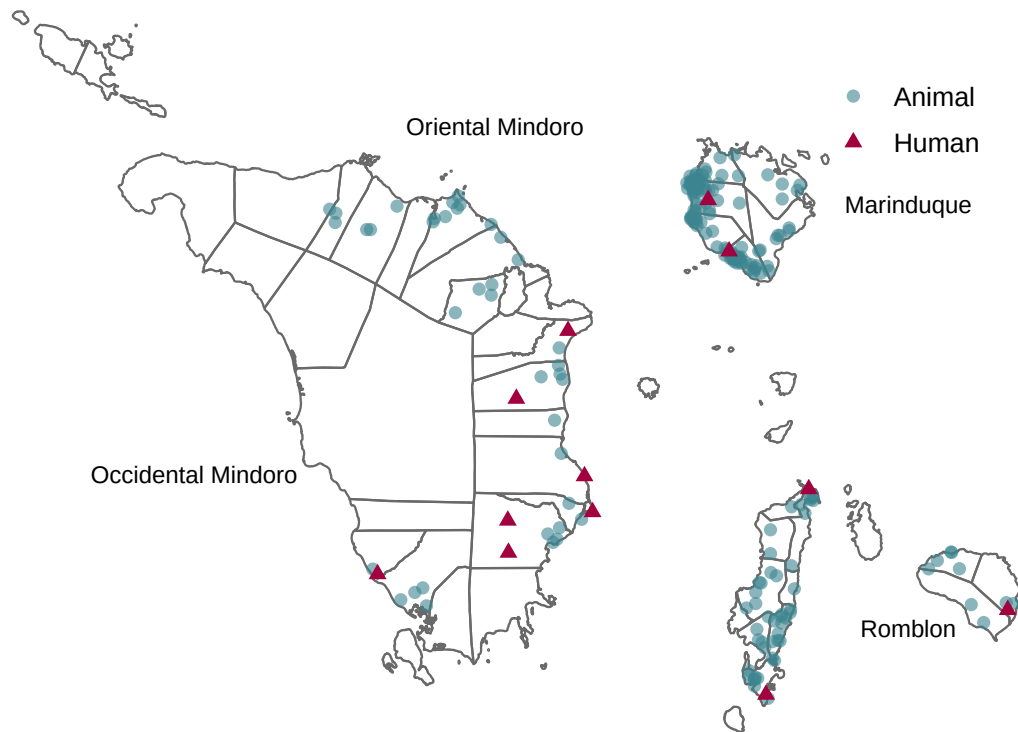
3.1 .1 Epidemic curve

Monthly reported animal rabies cases (confirmed and probable) and human rabies deaths in Oriental Mindoro, Romblon, Marinduque and Occidental Mindoro are presented in a time series.



3.2 .2 Locations of cases

The spatial distribution of all these cases are mapped. Human deaths are shown as triangles, whilst animal cases are shown as circles.



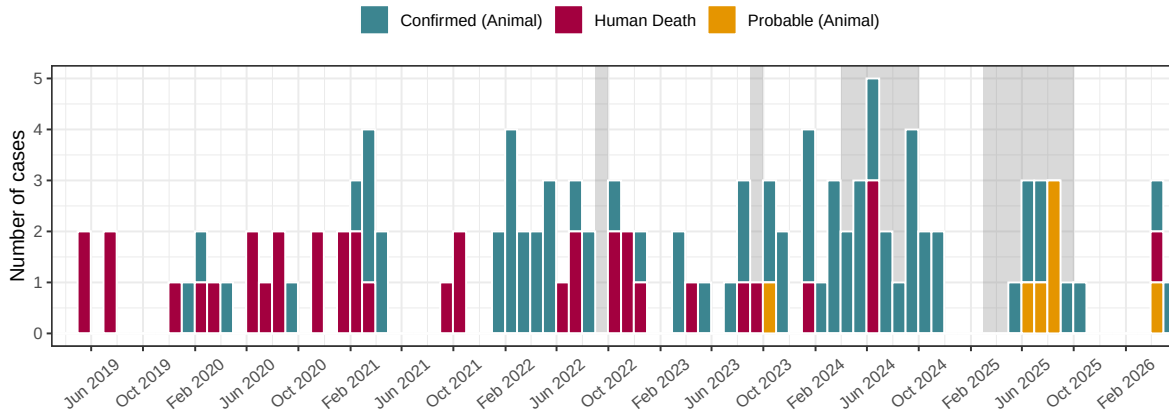
4 How did the epidemiological situation evolve in each province?

Looking at the epidemic curve by province reveals whether provinces were affected simultaneously or sequentially — this is important for understanding routes of disease spread and the timing of introduction events

4.1 Oriental Mindoro

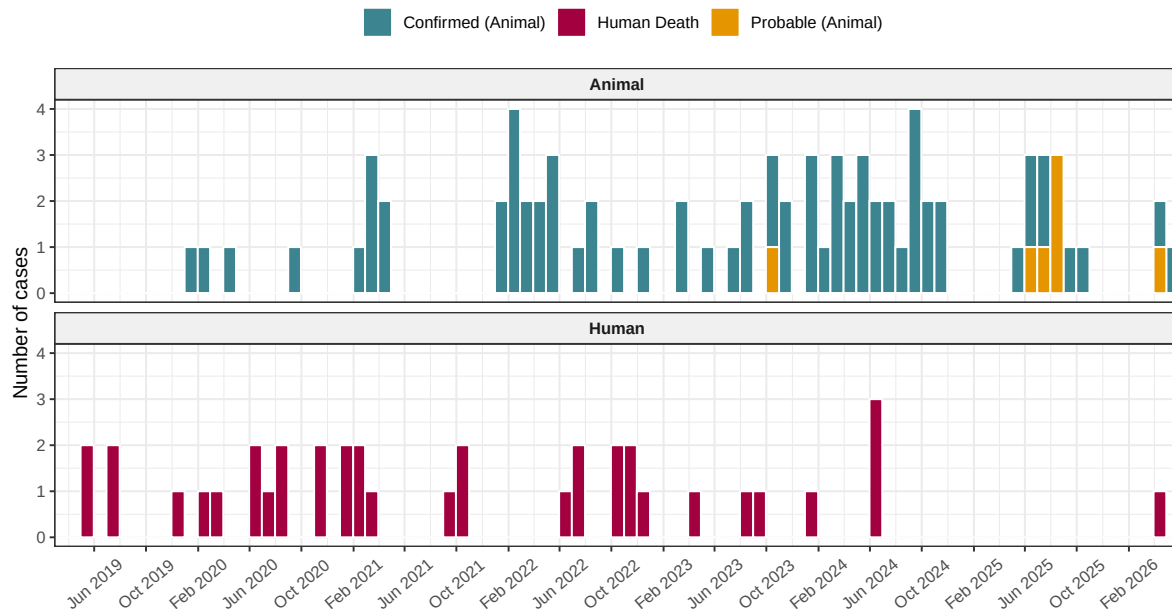
Key questions and things to explore:

- What is happening in the early outbreak period?
- How do animal cases compare to human cases in Oriental Mindoro?
- Roughly 1 person is bitten every 3 rabid dogs. Of people bitten, 90% get PEP. Of the 10% that don't get PEP, 20% will develop rabies and die. How does this change your interpretation of the early outbreak period?
- How does the most recent human death in April 2026 (look at the 2026 map!) change your interpretation of what is happening in Oriental Mindoro?

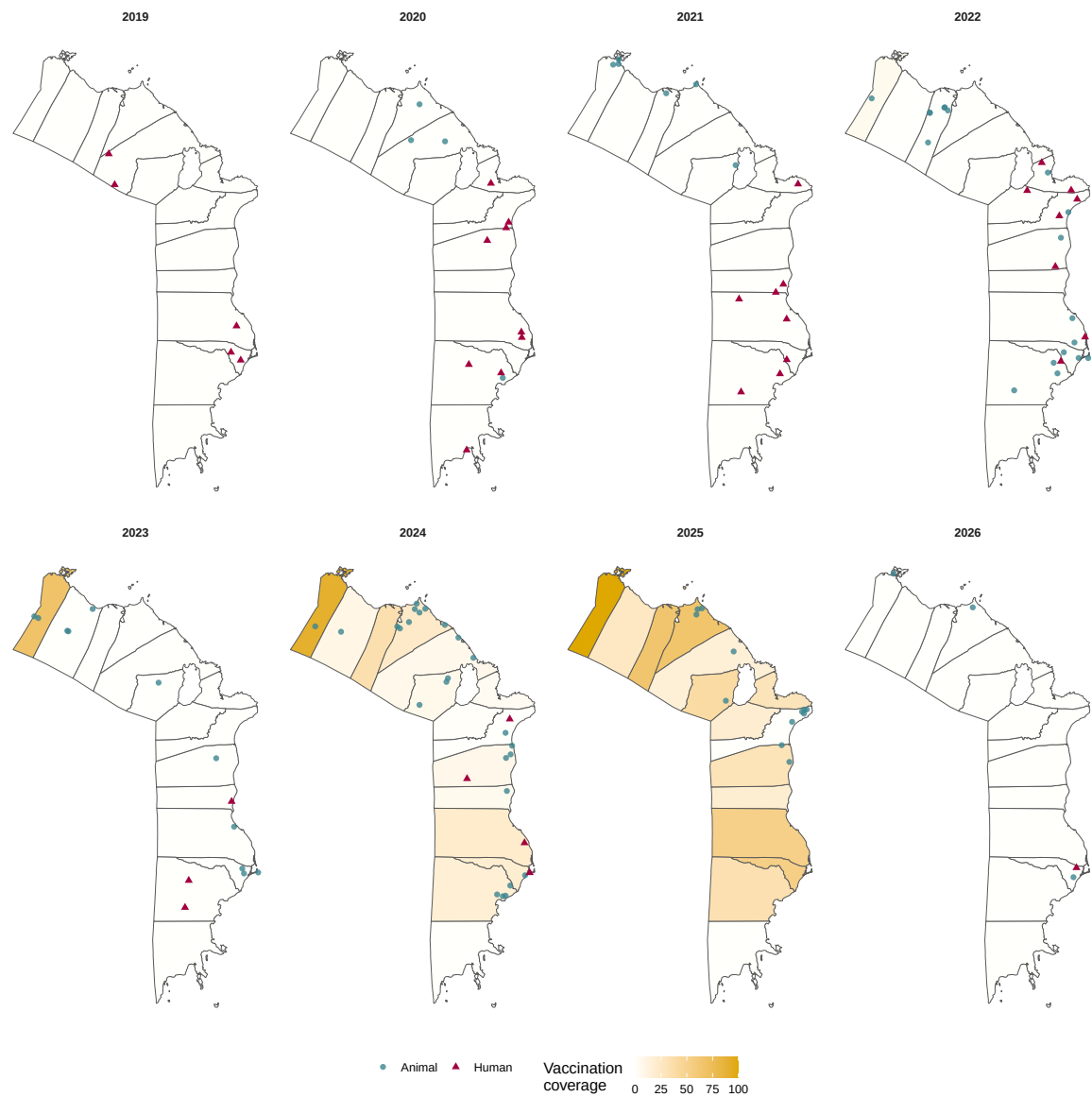


Dates of mass dog vaccination campaigns are shown in grey (a campaign is defined as >700 dogs vaccinated in a municipality in a month)

4.1.1 How do human cases compare to animal cases in Oriental Mindoro?



4.1.2 Does the distribution of cases over time and across municipalities suggest any gaps in surveillance (from the health or animal health perspective)?



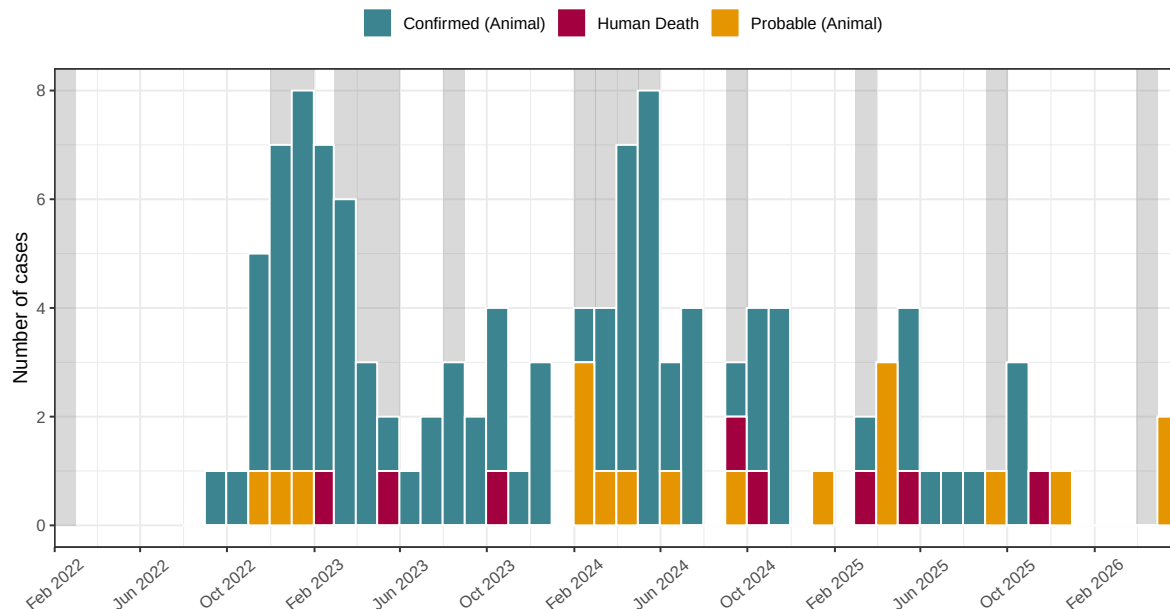
Note: some vaccination has occurred in 2026, however this data has not yet been released

4.2 Romblon

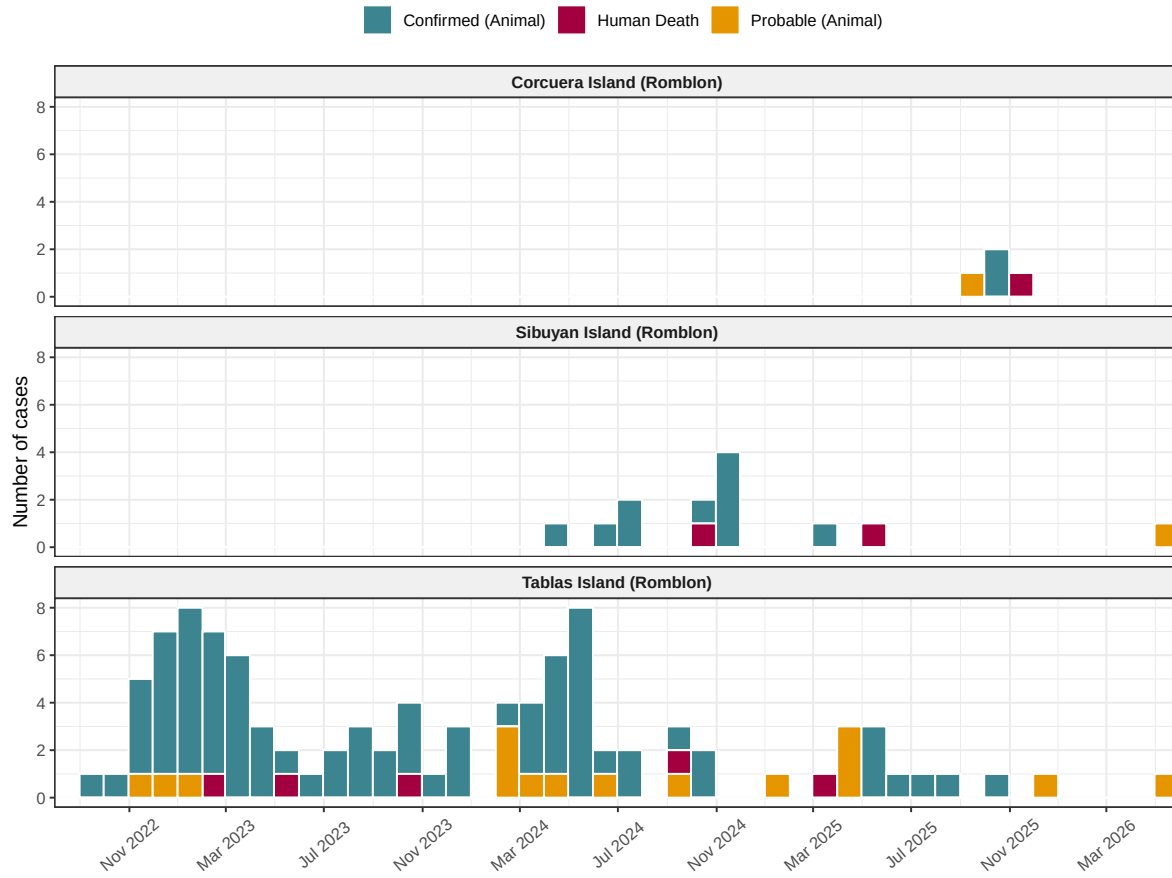
Key questions and things to consider:

- Romblon province is made up of different islands. How did rabies spread across the islands over time?
- If you were facing the situation in Tablas island in December 2022 (look at the 2022 map), what actions would you recommend to the local authorities?
- When did the first death occur? What can you say about the responsiveness of dog vaccination efforts?
- What if you were facing the situation in Sibuyan island in December 2023, would you be concerned? what communications would you want between municipalities? How might your thinking change by August 2024? Do you have any concerns if you are from Corcuera island? What do these complete plots tell you about the current surveillance efforts in Romblon?

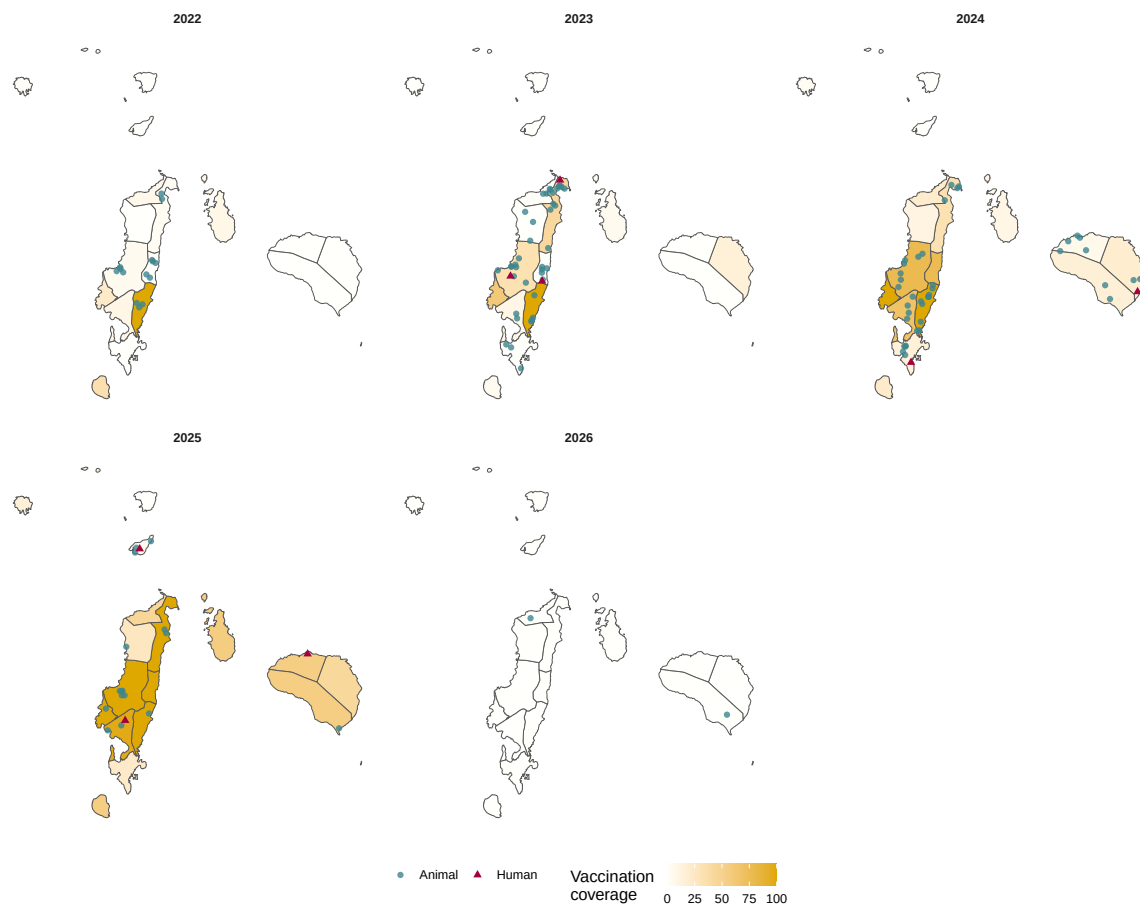
4.2.1 How does the outbreak look in Romblon if we break this down by island?



Dates of mass dog vaccination campaigns are shown in grey (MDV campaign defined as >700 dogs vaccinated in a municipality in a month)



4.2.2 Spatial distributions of human and animal rabies cases in Romblon by year



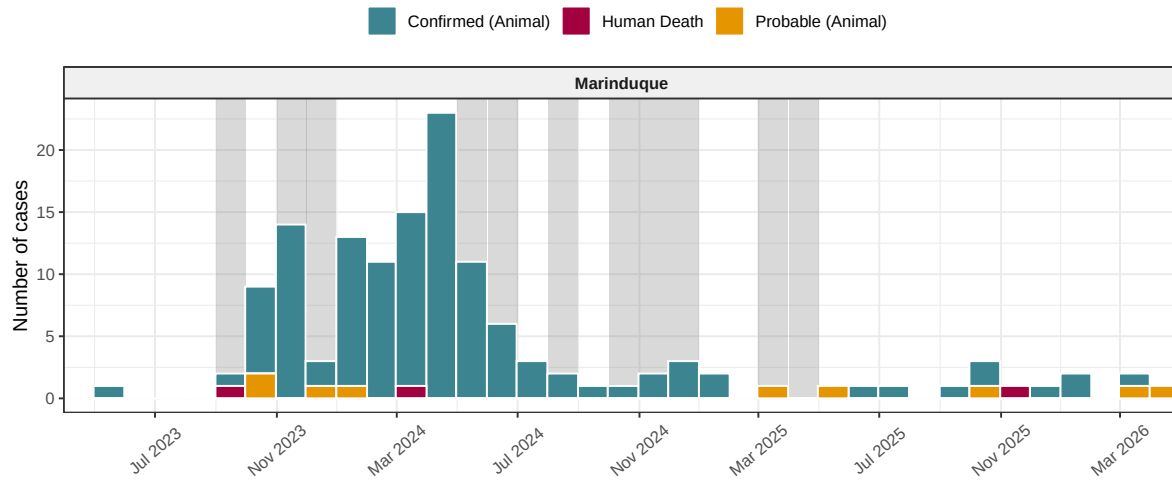
Note: some vaccination has occurred in 2026, however this data has not yet been released

4.3 Marinduque

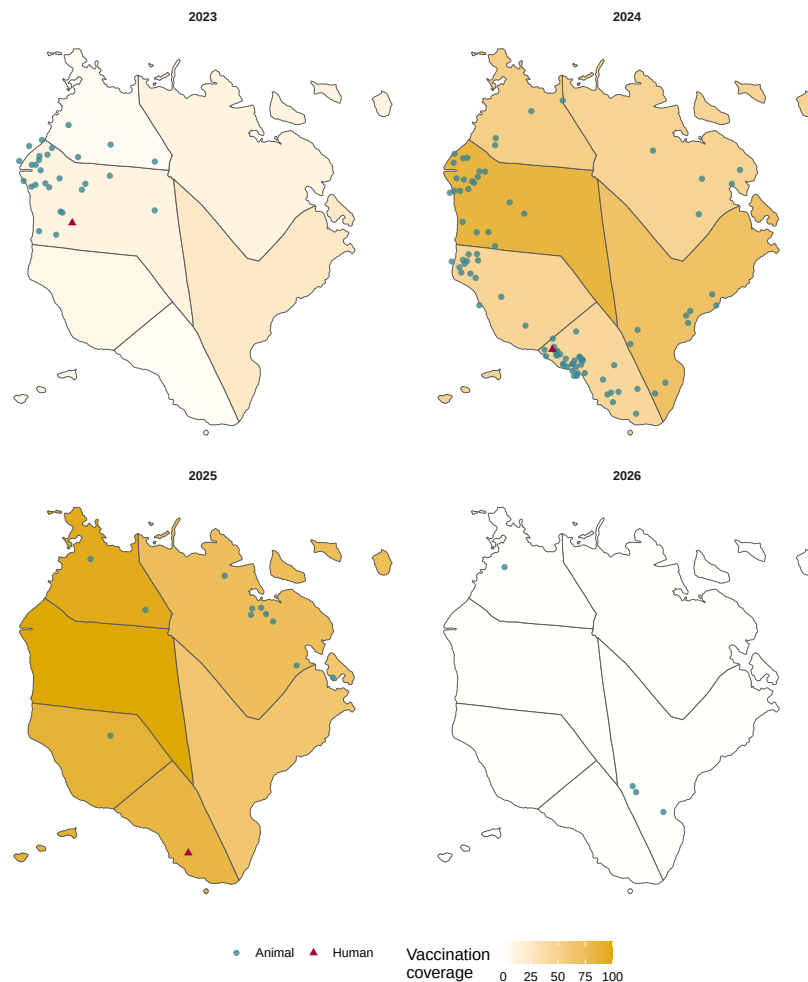
Key questions and things to consider:

- How did the outbreak evolve in Marinduque over time?
- What do these plots tell you about the current surveillance efforts in Marinduque?
- When did the first death occur? And how does this correspond to the increase in dog vaccination efforts?

- How does the outbreak in Marinduque compare to the outbreaks in Oriental mindoro and Romblon provinces?
- How does the human death in Dec 2025 impact your interpretation?



Dates of mass dog vaccination campaigns are shown in grey (a campaign defined as >700 dogs vaccinated in a municipality in a month)



Note: some vaccination has occurred in 2026, however this data has not yet been released

5 Providing the regional context

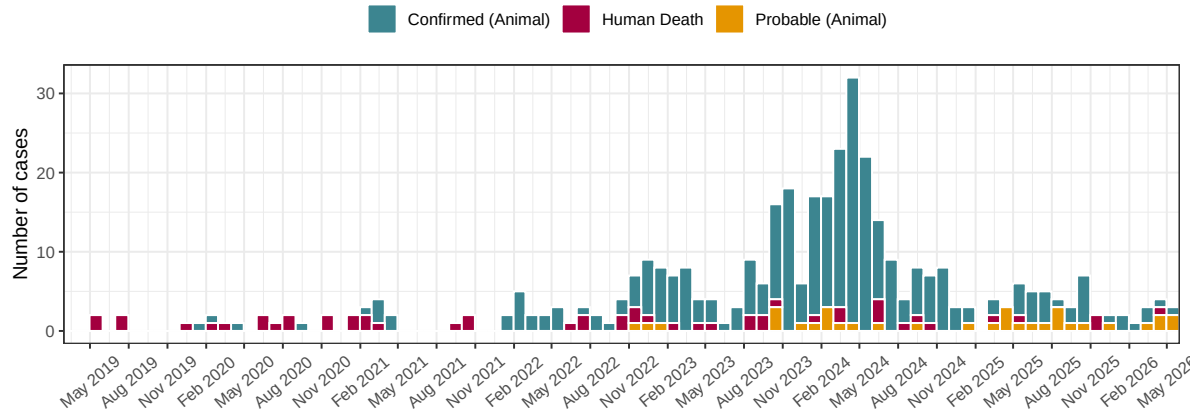
Key questions and things to explore:

- Which province do you think has the most robust surveillance? (consider surveillance from the veterinary perspective, the public health perspective and under a One Health approach)
- Do you consider rabies to be endemic in any of the provinces?
- What do you think are the triggers of outbreak responses?

- Why do the ratios of human to animal cases differ in the three provinces?

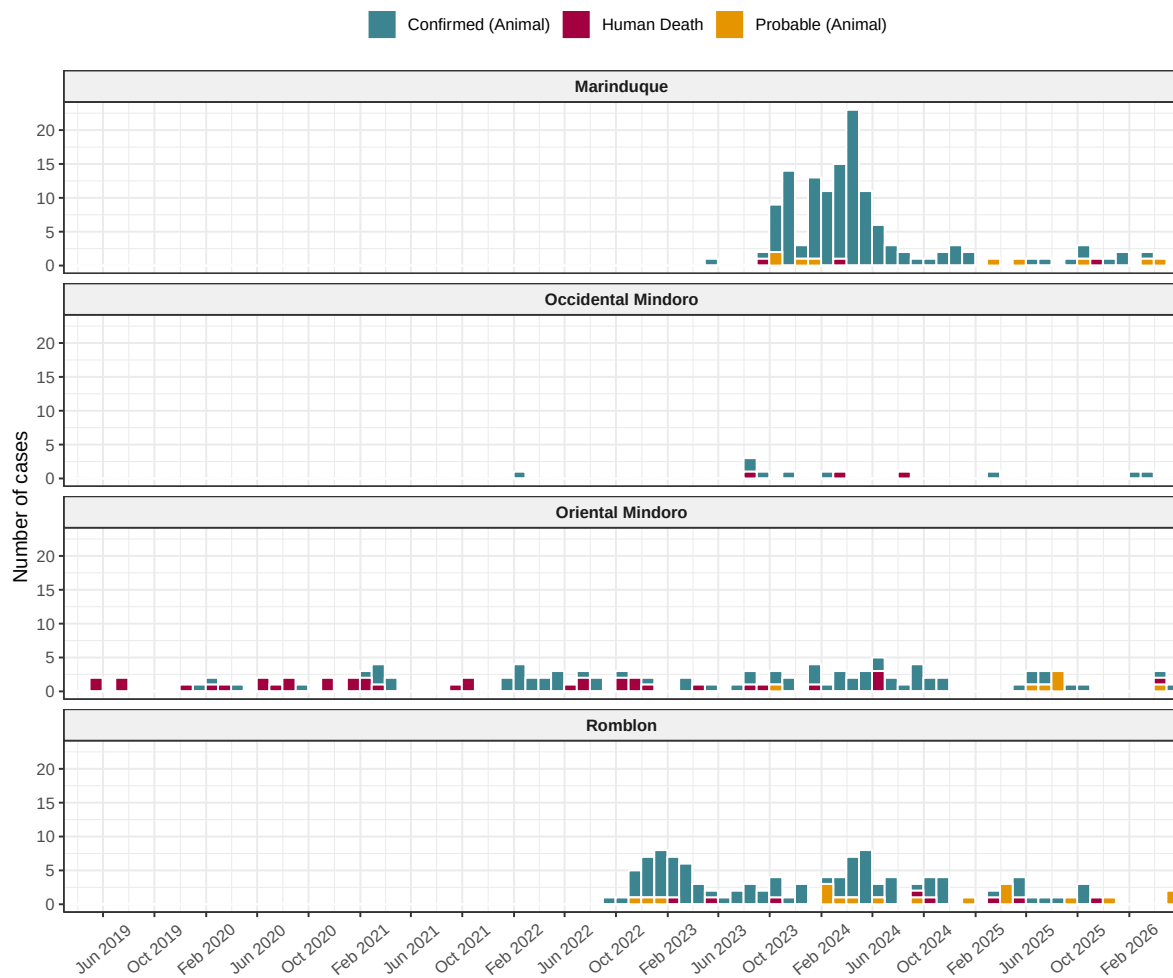
5.1 Extended epidemic curve

Monthly reported animal cases (confirmed and probable) and human deaths in Oriental Mindoro, Romblon, Marinduque and Occidental Mindoro including all data from 2019 - 2026



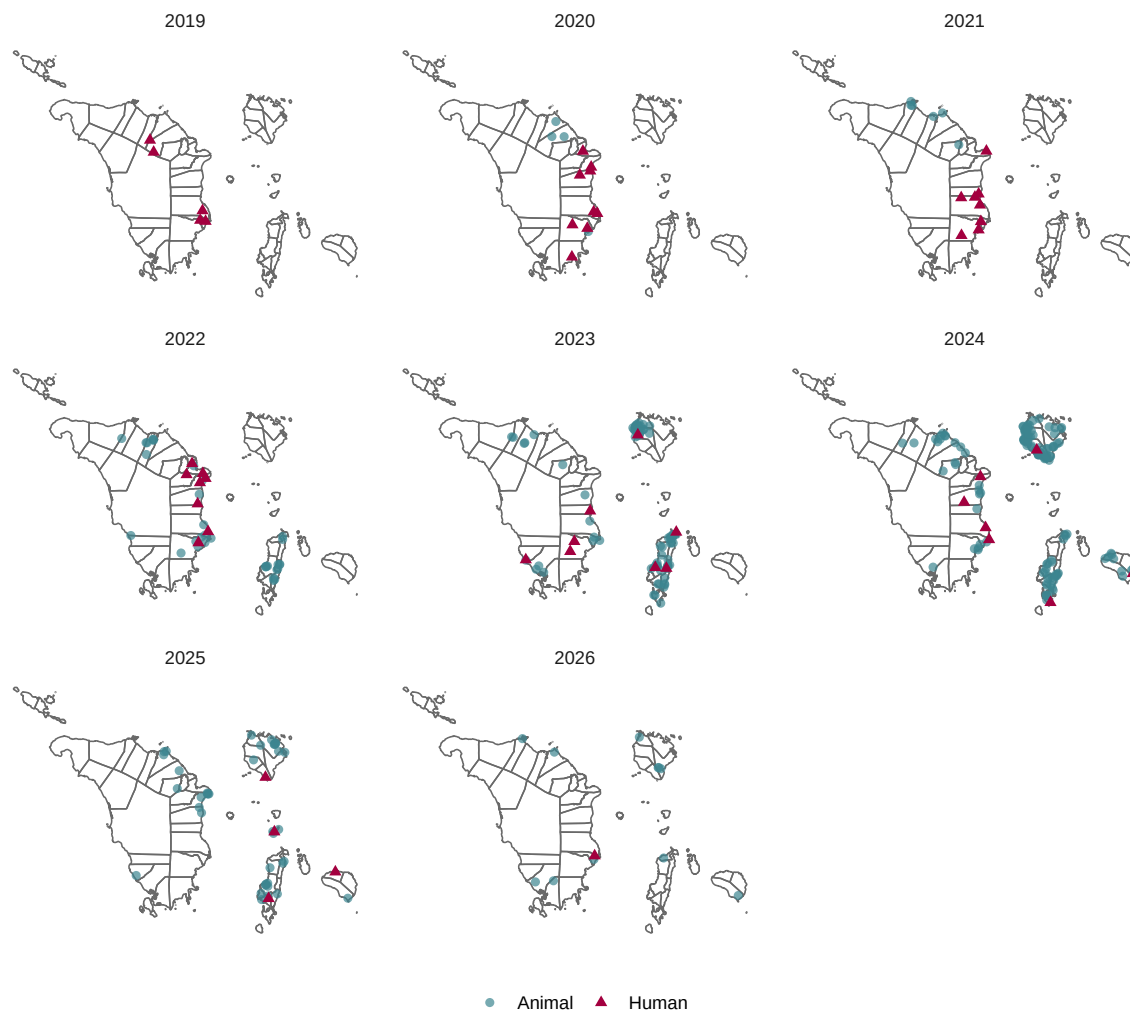
5.2 Epidemic curve by province

Reported rabies cases by province



5.3 Regional location of cases

Spatial distribution of all cases from 2019 to 2026. Human deaths are shown as triangles, whilst animal cases are shown as circles.



5.4 How do human deaths compare with animal cases?

A critical question is whether surveillance is sufficient. If we have areas where human deaths occur with relatively few reported animal cases this would suggest under-reporting of animal cases. Therefore, the annual incidence and the ratio of human deaths to animal cases may provide an indication of the surveillance sensitivity and reporting.

5.4.1 Ratio of human deaths to animal cases

How do the ratios of human deaths to animal cases vary across provinces? What do these suggest about the surveillance strength?

Province	Animal cases	Human deaths	Deaths per 100 animal cases
Marinduque	134	3	2.2
Occidental Mindoro	9	3	33.3
Oriental Mindoro	79	38	48.1
Romblon	110	8	7.3

5.5 Is rabies an emerging infectious disease threat or a neglected disease of poverty?

Rabies can be viewed as an emerging disease that can cause outbreaks in areas where it was previously absent. But it is also a disease that is entirely vaccine-preventable, that can be eliminated through dog vaccination, whilst human deaths only occur if people at-risk do not access emergency life-saving vaccines after an exposure.

The last decade in MIMAROPA region illustrates all of these different situations. Can you quantify the incidence of human rabies deaths and of rabies in dogs in each province and how this varies by year? What is the highest incidence? and what is the lowest incidence? Taking these extremes, and extrapolating to the entirety of the Philippines, what is the worst case scenario? how many rabies deaths *could* occur across the country if they were all to face the same challenges in rabies control and prevention?

And what if rabies control improved and surveillance was better - what is the best case scenario? What are the obstacles to achieving this best case outcome? and what can be done better that can support the best outcome?!

5.5.1 Population estimates by province

Location	Human population	Dog population
Oriental Mindoro	900,000	200,000
Occidental Mindoro	510,000	110,000
Marinduque	220,000	40,000
Romblon	284,000	55,800

5.5.2 Incidence of human deaths and animal cases

Calculating the annual incidence of human deaths and animal cases by province

Province	Animal cases	Animal incidence (per 1,000 dogs/year)	Human deaths	Human incidence (per 100,000 people/year)
Marinduque	134	0.48	3	0.19
Occidental Mindoro	9	0.01	3	0.08
Oriental Mindoro	79	0.06	38	0.59
Romblon	110	0.28	8	0.40
Overall	332	0.12	52	0.38

5.5.3 How does the annual incidence extrapolate?

Romblon in 2023 and 2025 recorded the highest annual human rabies incidence at 1.06 deaths per 100,000 people. What happens if we extrapolate these numbers to the overall population (population estimate of the Philippines ~113 million)?

Scenario	Rate	Expected_deaths
Regional average	0.38	429
Worst case	1.06	1198